Further Integration Techniques

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Finding the area between two curves using integration**

Find the area of the region enclosed between $y = \sin x$ and $y = \cos x$ for $\pi/4 \le x \le 5\pi/4$ (the interval on which $\sin x \ge \cos x$).

Area $= \int_{\pi/4}^{5\pi/4} (\sin x - \cos x)\,dx = [-\cos x - \sin x]_{\pi/4}^{5\pi/4}$. At $x = 5\pi/4$: $-\cos 5\pi/4 - \sin 5\pi/4 = \sqrt{2}/2 + \sqrt{2}/2 = \sqrt{2}$. At $x = \pi/4$: $-\cos \pi/4 - \sin \pi/4 = -\sqrt{2}/2 - \sqrt{2}/2 = -\sqrt{2}$. Area $= \sqrt{2} - (-\sqrt{2}) = 2\sqrt{2}$.

2 **[Foundational]** • **Target: Integration by substitution**

Evaluate $\int 2x(x^2 + 1)^4\,dx$ using the substitution $u = x^2 + 1$.

$du = 2x\,dx$. So $\int (x^2 + 1)^4 \cdot 2x\,dx = \int u^4\,du = u^5/5 + C = (x^2 + 1)^5/5 + C$.

3 **[Foundational]** • **Target: Integration by substitution**

Evaluate $\int_0^{\pi/2} \sin^3 x \cos x\,dx$ using the substitution $u = \sin x$.

$du = \cos x\,dx$. When $x = 0$, $u = 0$; when $x = \pi/2$, $u = 1$. $\int_0^1 u^3\,du = [u^4/4]_0^1 = 1/4$.

4 **[Foundational]** • **Target: Integration by parts**

Evaluate $\int xe^x\,dx$ using integration by parts.

Let $u = x$, $dv = e^x\,dx$, so $du = dx$, $v = e^x$. $\int xe^x\,dx = xe^x - \int e^x\,dx = xe^x - e^x + C = (x - 1)e^x + C$.

5 **[Foundational]** • **Target: Integration by parts**

Evaluate $\int x \ln x\,dx$ using integration by parts.

Let $u = \ln x$, $dv = x\,dx$, so $du = (1/x)\,dx$, $v = x^2/2$. $\int x \ln x\,dx = x^2/2 \ln x - \int x^2/2 \cdot 1/x\,dx = x^2/2 \ln x - x^2/4 + C$.

6 **[Foundational]** • **Target: Repeated integration by parts**

Evaluate $\int x^2 e^x\,dx$, applying integration by parts twice.

First application: $u = x^2$, $dv = e^x dx$: $\int x^2 e^x\,dx = x^2 e^x - \int 2xe^x\,dx$. From the previous problem, $\int xe^x\,dx = (x - 1)e^x + C$, so $\int 2xe^x\,dx = 2(x - 1)e^x$. Result: $x^2 e^x - 2(x - 1)e^x + C = (x^2 - 2x + 2)e^x + C$.

7 **[Foundational]** • **Target: Integration using trigonometric identities**

Evaluate $\int \cos^2 x\,dx$ using the identity $\cos^2 x = (1 + \cos 2x)/2$.

$\int (1 + \cos 2x)/2\,dx = 1/2(x + \sin 2x/2) + C = x/2 + \sin 2x/4 + C$.

© ClipSAT — clipsat.org | All Rights Reserved

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Integration using partial fractions**

Evaluate $\int 1/(x^2-1)\,dx$ by first expressing the integrand in partial fractions.

$1/(x^2-1) = 1/((x-1)(x+1)) = A/(x-1) + B/(x+1)$. Solving: $A = 1/2$, $B = -1/2$.

$\int (1/2/(x-1) - 1/2/(x+1))dx = 1/2\ln|x-1| - 1/2\ln|x+1| + C = 1/2\ln|(x-1)/(x+1)| + C$.

9 **[Intermediate]** • **Target: Evaluating a definite integral**

Evaluate $\int_1^e \ln x/x\,dx$, using the substitution $u = \ln x$.

$du = (1/x)\,dx$. When $x = 1$, $u = 0$; when $x = e$, $u = 1$. $\int_0^1 u\,du = [u^2/2]_0^1 = 1/2$.

10 **[Intermediate]** • **Target: Evaluating a definite integral**

Evaluate $\int_0^\pi x\sin x\,dx$ using integration by parts.

Let $u = x$, $dv = \sin x\,dx$, so $du = dx$, $v = -\cos x$. $\int x\sin x\,dx = -x\cos x + \int \cos x\,dx = -x\cos x + \sin x$.
Evaluating from 0 to π: $[-\pi\cos\pi + \sin\pi] - [0 + 0] = -\pi(-1) + 0 = \pi$.

11 **[Intermediate]** • **Target: Integration involving inverse trigonometric results**

Evaluate $\int 1/(1+x^2)\,dx$, and use it to evaluate $\int_0^1 1/(1+x^2)\,dx$.

$\int 1/(1+x^2)\,dx = \arctan x + C$. So $\int_0^1 1/(1+x^2)\,dx = \arctan(1) - \arctan(0) = \pi/4 - 0 = \pi/4$.

12 **[Intermediate]** • **Target: Integration by substitution with trigonometric functions**

Evaluate $\int \sin^3 x\,dx$, using the identity $\sin^2 x = 1 - \cos^2 x$ and the substitution $u = \cos x$.

Rewrite: $\sin^3 x = \sin x(1 - \cos^2 x)$. Let $u = \cos x$, $du = -\sin x\,dx$.

$\int \sin x(1-\cos^2 x)\,dx = -\int (1-u^2)\,du = -u + u^3/3 + C = -\cos x + \cos^3 x/3 + C$.

13 **[Intermediate]** • **Target: Integration by substitution with trigonometric functions**

Evaluate $\int \tan x\,dx$, using the substitution $u = \cos x$.

$\tan x = \sin x/(\cos x)$. Let $u = \cos x$, $du = -\sin x\,dx$. $\int \sin x/(\cos x)\,dx = -\int 1/u\,du = -\ln|u| + C = -\ln|\cos x| + C = \ln|\sec x| + C$.

14 **[Intermediate]** • **Target: Integration by parts with a logarithmic integrand**

Evaluate $\int_1^e \ln x\,dx$ using integration by parts.

Let $u = \ln x$, $dv = dx$, so $du = (1/x)\,dx$, $v = x$. $\int \ln x\,dx = x\ln x - \int x \cdot 1/x\,dx = x\ln x - x + C$.
Evaluating from 1 to e: $[e\ln e - e] - [1\ln 1 - 1] = (e - e) - (0 - 1) = 0 + 1 = 1$.

© ClipSAT — clipsat.org | All Rights Reserved

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Integration using a Weierstrass-style substitution** • **Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)**

Evaluate $\int 1/(5+4\cos x)\,dx$ using the substitution $t=\tan(x/2)$, given that this substitution transforms $\cos x=(1-t^2)/(1+t^2)$ and $dx=2/(1+t^2)dt$, leading to the simplified integrand $2/(9+t^2)$.

Substituting: $\int 1/(5+4(1-t^2)/(1+t^2))\cdot 2/(1+t^2)\,dt=\int 2/(5(1+t^2)+4(1-t^2))\,dt=\int 2/(9+t^2)\,dt$. Using the standard result $\int 1/(a^2+t^2)\,dt=(1/a)\arctan(t/a)+C$ with $a=3$: $2\cdot(1/3)\arctan(t/3)+C=(2/3)\arctan((1/3)\tan(x/2))+C$.

16 **[Challenging]** • **Target: Integration by parts with a trigonometric-exponential product** • **Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals**

Evaluate $\int e^x\cos x\,dx$, applying integration by parts twice and solving the resulting equation for the original integral.

Let $I=\int e^x\cos x\,dx$. First application: $u=\cos x$, $dv=e^x dx$, so $du=-\sin x\,dx$, $v=e^x$: $I=e^x\cos x+\int e^x\sin x\,dx$. Second application on $\int e^x\sin x\,dx$: $u=\sin x$, $dv=e^x dx$: $\int e^x\sin x\,dx=e^x\sin x-\int e^x\cos x\,dx=e^x\sin x-I$. Substituting back: $I=e^x\cos x+e^x\sin x-I$, so $2I=e^x(\cos x+\sin x)$, giving $I=e^x(\cos x+\sin x)/2+C$.

17 **[Challenging]** • **Target: Finding the area between two curves using integration** • **Cross-Lesson: 4.02 The Poisson Distribution**

Evaluate $\int x^2\ln x\,dx$ using integration by parts.

Let $u=\ln x$, $dv=x^2dx$, so $du=(1/x)\,dx$, $v=x^3/3$. $\int x^2\ln x\,dx=x^3/3\ln x-\int x^3/3\cdot 1/x\,dx=x^3/3\ln x-\int x^2/3\,dx=x^3/3\ln x-x^3/9+C$.

18 **[Challenging]** • **Target: Finding the area between two curves using integration** • **Cross-Lesson: 5.02 Differential Equations**

A particle's velocity is given by $v(t)=t\cos t$ metres per second. Find the particle's displacement over the interval $0\le t\le\pi$, using integration by parts.

Displacement $=\int_0^{\pi} t\cos t\,dt$. Using integration by parts with $u=t$, $dv=\cos t\,dt$, so $du=dt$, $v=\sin t$: $\int t\cos t\,dt=t\sin t-\int\sin t\,dt=t\sin t+\cos t+C$. Evaluating from 0 to π: $[\pi\sin\pi+\cos\pi]-[0\sin 0+\cos 0]=[0-1]-[0+1]=-1-1=-2$. Displacement $=-2$ metres.

19 **[Challenging]** • **Target: Finding the area between two curves using integration** • **Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series**

This question has three parts. (a) Use the substitution $u=x^2+4$ to find $\int x/(x^2+4)\,dx$. (b) Use integration by parts to find $\int\arctan x\,dx$. (c) Hence, using the result of part (b) and the standard result $\int 1/(1+x^2)dx=\arctan x+C$, evaluate $\int_0^1\arctan x\,dx$, giving an exact answer in terms of π and $\ln$.

(a) $u=x^2+4$, $du=2x\,dx$: $\int x/(x^2+4)\,dx=1/2\int 1/u\,du=1/2\ln|x^2+4|+C$. (b) Let $u=\arctan x$, $dv=dx$, so $du=(1/(1+x^2))\,dx$, $v=x$: $\int\arctan x\,dx=x\arctan x-\int x/(1+x^2)\,dx=x\arctan x-1/2\ln(1+x^2)+C$ (using the result from part (a), with 4 replaced by 1). (c) Evaluating from 0 to 1: $[1\cdot\arctan 1-1/2\ln 2]-[0-1/2\ln 1]=\pi/4-1/2\ln 2$.

© ClipSAT — clipsat.org | All Rights Reserved

20 [Challenging] • Target: Reduction formulas • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

Given the reduction formula $I_n = \int x^n e^x \, dx = x^n e^x - nI_{n-1}$ (obtained via integration by parts), and that $I_0 = e^x + C$, use it to find $I_2 = \int x^2 e^x \, dx$.

$I_1 = x^1 e^x - 1 \cdot I_0 = xe^x - e^x$ (ignoring the constant until the final step).

$I_2 = x^2 e^x - 2I_1 = x^2 e^x - 2(xe^x - e^x) = x^2 e^x - 2xe^x + 2e^x + C = (x^2 - 2x + 2)e^x + C$, matching the result found earlier in this worksheet by direct repeated integration by parts.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Choosing an appropriate integration technique

Which integration technique is most appropriate for evaluating $\int x^3 \sin x \, dx$?

a. Integration by parts, applied three times (reducing the power of x each time) ✓

b. Substitution $u = \sin x$

c. Partial fractions

d. The double-angle identity for $\sin$

Since the integrand is a product of a polynomial and a trigonometric function, integration by parts is appropriate — and because the polynomial factor is x^3 (not x^1), the process must be repeated three times, each time reducing the power of x by one, until the polynomial factor disappears.

22 [Foundational] • Target: Integration by Parts (Repeated & Cyclic)

Evaluate $\int x^2 \cos x dx$ using integration by parts.

a. $x^2 \sin x + 2x \cos x - 2\sin x + C$ ✓

b. $x^2 \sin x - 2x \cos x + 2\sin x + C$

c. $-x^2 \sin x + 2x \cos x + C$

d. $\frac{1}{3}x^3 \sin x + C$

Tabular: $x^2 \to 2x \to 2 \to 0$; $\cos x \to \sin x \to -\cos x \to -\sin x$. Alternating:

$+x^2 \sin x - 2x(-\cos x) + 2(-\sin x) = x^2 \sin x + 2x \cos x - 2\sin x + C$.

© ClipSAT — clipsat.org | All Rights Reserved

23 [Foundational] • Target: Integration by Parts (Repeated & Cyclic)

Evaluate the cyclic integral $I = \int e^x \sin x\,dx$.

a. $\frac{1}{2}e^x(\sin x - \cos x) + C$ ✓

b. $\frac{1}{2}e^x(\sin x + \cos x) + C$

c. $e^x(\sin x - \cos x) + C$

d. $-e^x \cos x + C$

Integrate by parts twice: $I = e^x \sin x - \int e^x \cos x\,dx = e^x \sin x - [e^x \cos x - \int e^x(-\sin x)\,dx] = e^x(\sin x - \cos x) - I \Rightarrow 2I = e^x(\sin x - \cos x) \Rightarrow I = \frac{1}{2}e^x(\sin x - \cos x) + C$.

24 [Foundational] • Target: Integration by Parts (Repeated & Cyclic)

Evaluate $\int_1^e \ln x\,dx$.

a. 1 ✓

b. $e - 1$

c. e

d. 0

$\int \ln x\,dx = x\ln x - x$. Evaluating from 1 to e: $[e\ln e - e] - [1\ln 1 - 1] = [e - e] - [0 - 1] = 0 - (-1) = 1$.

25 [Foundational] • Target: Trigonometric Substitutions

Which trigonometric substitution is best suited to evaluate $\int \frac{1}{\sqrt{16 - x^2}}dx$?

a. $x = 4\sin\theta$ ✓

b. $x = 4\tan\theta$

c. $x = 4\sec\theta$

d. $x = 16\sin\theta$

For forms $\sqrt{a^2 - x^2}$, the substitution $x = a\sin\theta$ simplifies the radical via $16 - 16\sin^2\theta = 16\cos^2\theta$.

26 [Foundational] • Target: Trigonometric Substitutions

Evaluate $\int \frac{1}{x^2 + 16}dx$.

a. $\frac{1}{4}\arctan\left(\frac{x}{4}\right) + C$ ✓

b. $\arctan\left(\frac{x}{4}\right) + C$

c. $\frac{1}{16}\arctan\left(\frac{x}{16}\right) + C$

d. $\frac{1}{4}\ln|x^2 + 16| + C$

Standard formula $\int \frac{1}{x^2 + a^2}dx = \frac{1}{a}\arctan\left(\frac{x}{a}\right) + C$. With $a = 4$: $\frac{1}{4}\arctan\left(\frac{x}{4}\right) + C$.

© ClipSAT — clipsat.org | All Rights Reserved

27 [Foundational] • Target: Trigonometric Substitutions

Which substitution evaluates $\int \sqrt{x^2 - 9}\,dx$?

a. $x = 3\sec\theta$ ✓

b. $x = 3\sin\theta$

c. $x = 3\tan\theta$

d. $x = 9\sec\theta$

For $\sqrt{x^2 - a^2}$, use $x = a\sec\theta$, since $x^2 - 9 = 9(\sec^2\theta - 1) = 9\tan^2\theta$.

28 [Foundational] • Target: Integrals of Powers of Trigonometric Functions

Evaluate $\int \sin^3 x dx$.

a. $-\cos x + \frac{1}{3}\cos^3 x + C$ ✓

b. $\cos x - \frac{1}{3}\cos^3 x + C$

c. $\frac{1}{4}\sin^4 x + C$

d. $-\frac{1}{3}\cos^3 x + C$

Rewrite $\sin^3 x = (1 - \cos^2 x)\sin x$. Let $u = \cos x$, $du = -\sin x dx$:
$\int -(1 - u^2)du = -u + \frac{u^3}{3} + C = -\cos x + \frac{1}{3}\cos^3 x + C$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Integrals of Powers of Trigonometric Functions

Evaluate $\int \cos^2 x dx$.

a. $\frac{1}{2}x + \frac{1}{4}\sin(2x) + C$ ✓

b. $\frac{1}{2}x - \frac{1}{4}\sin(2x) + C$

c. $\frac{1}{3}\cos^3 x + C$

d. $\frac{1}{2}x + \sin(2x) + C$

Use double angle identity $\cos^2 x = \frac{1 + \cos(2x)}{2}$: $\int \frac{1 + \cos(2x)}{2} dx = \frac{x}{2} + \frac{\sin(2x)}{4} + C$.

© ClipSAT — clipsat.org | All Rights Reserved

30 [Intermediate] • Target: Integrals of Powers of Trigonometric Functions

Evaluate $\int \tan^2 x\,dx$.

a. $\tan x - x + C$ ✓

b. $\sec^2 x + C$

c. $\tan x + x + C$

d. $\frac{1}{3}\tan^3 x + C$

Identity: $\tan^2 x = \sec^2 x - 1$. Integrating: $\int (\sec^2 x - 1)dx = \tan x - x + C$.

31 [Intermediate] • Target: Integration by Substitution (Advanced Forms)

Evaluate $\int \frac{x}{\sqrt{x+1}}dx$ using substitution $u = x + 1$.

a. $\frac{2}{3}(x+1)^{3/2} - 2(x+1)^{1/2} + C$ ✓

b. $\frac{2}{3}(x+1)^{3/2} + 2(x+1)^{1/2} + C$

c. $(x+1)^{3/2} - (x+1)^{1/2} + C$

d. $\frac{1}{2}(x+1)^2 + C$

With $u = x + 1$, $x = u - 1$ and $dx = du$: $\int \frac{u-1}{u^{1/2}}du = \int (u^{1/2} - u^{-1/2})du = \frac{2}{3}u^{3/2} - 2u^{1/2} + C = \frac{2}{3}(x+1)^{3/2} - 2(x+1)^{1/2} + C$.

32 [Intermediate] • Target: Integration by Substitution (Advanced Forms)

Evaluate $\int_0^1 \frac{e^x}{1+e^x}dx$.

a. $\ln\left(\frac{1+e}{2}\right)$ ✓

b. $\ln(1+e)$

c. $e - 1$

d. $\frac{e}{2}$

Let $u = 1 + e^x$, $du = e^x dx$. At $x = 0$, $u = 2$; at $x = 1$, $u = 1 + e$.
$\int_2^{1+e} \frac{1}{u}du = [\ln|u|]_2^{1+e} = \ln(1+e) - \ln 2 = \ln\left(\frac{1+e}{2}\right)$.

33 [Intermediate] • Target: Integration by Substitution (Advanced Forms)

Evaluate $\int \frac{1}{x\ln x}dx$ for $x > 1$.

a. $\ln|\ln x| + C$ ✓

b. $\frac{1}{2}(\ln x)^2 + C$

c. $\frac{1}{\ln x} + C$

d. $x\ln x - x + C$

Let $u = \ln x$, $du = \frac{1}{x}dx$. $\int \frac{1}{u}du = \ln|u| + C = \ln|\ln x| + C$.

© ClipSAT — clipsat.org | All Rights Reserved

34 [Intermediate] • Target: Integration by substitution with a linear inner function

Consider the function $f(x) = \frac{\sin(2x)}{\cos(x)}$. For the domain $x \in [0, 2\pi]$ such that $\cos(x) eq 0$, find the exact value of the definite integral $\int_0^{\pi/2} f(x)dx$.

a. $\ln(2)$ **b.** 1

c. 2 ✓ **d.** $\ln(1)$

First, we simplify the function $f(x)$. Using the double angle identity for sine, $\sin(2x) = 2\sin(x)\cos(x)$. Therefore, $f(x) = \frac{2\sin(x)\cos(x)}{\cos(x)}$. For $\cos(x) eq 0$, we can cancel the $\cos(x)$ terms, so $f(x) = 2\sin(x)$. Now we need to evaluate the definite integral: $\int_0^{\pi/2} 2\sin(x)dx$. The antiderivative of $2\sin(x)$ is $-2\cos(x)$. Evaluating the definite integral using the Fundamental Theorem of Calculus: $[-2\cos(x)]_0^{\pi/2} = -2\cos(\pi/2) - (-2\cos(0))$.

35 [Intermediate] • Target: Integration using the sine double-angle identity

If $f(x) = \frac{7}{x}$ (a self-inverse function), find the positive value of x for which $f(x) = x$ (a fixed point).

a. $\sqrt{7}$ ✓ **b.** $\sqrt{10}$

c. $\sqrt{5}$ **d.** $\sqrt{6}$

$\frac{7}{x} = x \Rightarrow x^2 = 7 \Rightarrow x = \sqrt{7}$

36 [Intermediate] • Target: Volume of revolution using integration

If $f(x) = \frac{8}{x}$ (a self-inverse function), find the positive value of x for which $f(x) = x$ (a fixed point).

a. 3 **b.** $\sqrt{8}$ ✓

c. $\sqrt{11}$ **d.** $\sqrt{6}$

$\frac{8}{x} = x \Rightarrow x^2 = 8 \Rightarrow x = \sqrt{8}$

37 [Intermediate] • Target: Substitution with a shifting definite integral bound

If $f(x) = \frac{5}{x}$ (a self-inverse function), find the positive value of x for which $f(x) = x$ (a fixed point).

a. $\sqrt{8}$ **b.** 2

c. $\sqrt{5}$ ✓ **d.** $\sqrt{7}$

$\frac{5}{x} = x \Rightarrow x^2 = 5 \Rightarrow x = \sqrt{5}$

© ClipSAT — clipsat.org | All Rights Reserved

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Choosing an appropriate integration technique • Cross-Lesson: 5.02 Differential Equations

If $f(x) = \frac{3}{x}$ (a self-inverse function), find the positive value of x for which $f(x) = x$ (a fixed point).

a. $\sqrt{3}$ ✓

b. $\sqrt{5}$

c. 2

d. $\sqrt{6}$

$\frac{3}{x} = x \Rightarrow x^2 = 3 \Rightarrow x = \sqrt{3}$

39 [Challenging] • Target: Integration by Parts (Repeated & Cyclic) • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

If $f(x) = \frac{4}{x}$ (a self-inverse function), find the positive value of x for which $f(x) = x$ (a fixed point).

a. **2** ✓

b. 1

c. 3

d. 4

$\frac{4}{x} = x \Rightarrow x^2 = 4 \Rightarrow x = 2$

40 [Challenging] • Target: Trigonometric Substitutions • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

Integration by parts chooses u such that:

a. u is simple to integrate

b. **du/dx is simpler than u** ✓

c. dv is the whole integrand

d. u contains e^x always

Choose u so that du/dx simplifies the remaining integral (LIATE rule).

41 [Challenging] • Target: Integrals of Powers of Trigonometric Functions • Cross-Lesson: 6.01 Matrices (AI HL)

Find the angle between lines with direction vectors $\mathbf{d_1} = (1; 4; 8)$ and $\mathbf{d_2} = (4; -1; 0)$.

a. 90° ✓

b. 45°

c. 60°

d. 30°

$\mathbf{d_1} \cdot \mathbf{d_2} = 0 = 0 \Rightarrow \theta = 90^\circ$

© ClipSAT — clipsat.org | All Rights Reserved

42 [Challenging] • Target: Integration by Substitution (Advanced Forms) • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

Find the angle between lines with direction vectors $\mathbf{d_1} = (4; 4; 7)$ and $\mathbf{d_2} = (4; -4; 0)$.

a. 45°

b. 30°

c. 60°

d. 90° ✓

$\mathbf{d_1} \cdot \mathbf{d_2} = 0 = 0 \Rightarrow \theta = 90^\circ$

43 [Challenging] • Target: Integration by substitution • Cross-Lesson: 6.03 Markov Chains (AI HL)

Find the angle between lines with direction vectors $\mathbf{d_1} = (1; 2; 2)$ and $\mathbf{d_2} = (2; -1; 0)$.

a. 60°

b. 90° ✓

c. 45°

d. 30°

$\mathbf{d_1} \cdot \mathbf{d_2} = 0 = 0 \Rightarrow \theta = 90^\circ$

44 [Challenging] • Target: Integration by parts • Cross-Lesson: 1.01 Proof by Mathematical Induction

Find the angle between lines with direction vectors $\mathbf{d_1} = (3; 4; 12)$ and $\mathbf{d_2} = (4; -3; 0)$.

a. 90° ✓

b. 60°

c. 30°

d. 45°

$\mathbf{d_1} \cdot \mathbf{d_2} = 0 = 0 \Rightarrow \theta = 90^\circ$

45 [Challenging] • Target: Repeated integration by parts • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

Find the angle between lines with direction vectors $\mathbf{d_1} = (6; 2; 3)$ and $\mathbf{d_2} = (2; -6; 0)$.

a. 90° ✓

b. 45°

c. 60°

d. 30°

$\mathbf{d_1} \cdot \mathbf{d_2} = 0 = 0 \Rightarrow \theta = 90^\circ$

© ClipSAT — clipsat.org | All Rights Reserved